

3D Printing, Community Meets and Budget-Friendly Tools Drive Open Electronics

Open hardware is leading us to the next industrial revolution where digital fabrication (read 3D printers) is allowing everyone with an idea to bring it to fruition. Community meets, demo sessions and low-cost electronics are also fuelling open source electronics. There have been some new initiatives and innovations to give the open community a boost

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Open source electronics is enabling people to realise their designs quickly. Designers and hobbyists are turning their ideas into reality through digital manufacturing. The power of product development today is not only with the major corporations but is trickling down to individual designers and entrepreneurs. As a matter of fact, semiconductor multinationals are gradually investing significant time and energy on this do-it-yourself (DIY) community. The availability of low-cost manufacturing tools and low-cost distribution on the Internet sites are levelling the playing field. So let's explore the driving factors for people ranging from hobbyists, DIYers, specialists/experts, professionals and even academicians to create and build open source systems.

It is all about making and sharing

"Open source electronics is electronics designed with the spirit similar to open source software. When more eyes look at open source electronics,

any possible defect will be identified quickly and, more important, new ideas to utilise the same hardware for different applications will emerge," says Dhaval Vasa, solution architect, eInfochips.

With open source electronics, everything is to be released publicly. "I have been working on

you built it so that any new person can have an accelerated learning instead of starting from scratch." When hardware is open, it enables you to modify it, better it and pull out something really great.

More and more people going open

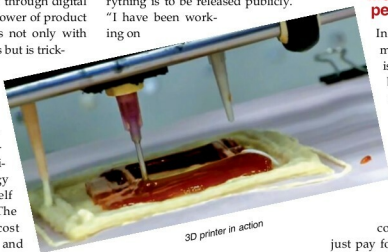
In India, the kind of development happening in the field is at an elementary stage, but there is a constant rise in the number of people exploring open source hardware. Kuber shares, "In 2009-10, when I started exploring open source hardware, there were not many people who had heard of such a

concept. The norm was you just pay for a course, learn 8051 and get back to make a summer project." Today, everyone is buying and building their own robotics project. People are widely adopting open source hardware. She adds, "In every technical session or robotics competition, there is at least one thing made using an Arduino hardware."

There are many people, especially those with software background, stepping up and creating open source products. Sivaram says, "But one thing I have noticed is, sometimes people try to achieve too many things in one project."

open source electronics for past many years and for me it is something non-proprietary—all documentation is made available, be it schematics or PCB layout," says Varun Sivaram, CEO and founder, Technigriti Systems Pvt Ltd. He adds, "Arduino, for instance, releases everything including the Eagle CAD file. Anyone can modify it and build stuff."

Open source electronics is all about sharing knowledge, believes Priya Kuber, director, Arduino India. She says, "It is not about just getting behind closed doors but teaching others how



3D printer in action